

Today's Learning Summary

HTML Tables & Images

1. HTML Tables

HTML tables are used to display data in rows and columns using the `<table>` tag. Today we practiced creating several types of tables.

Key Table Tags

Tag	Purpose
<code><table></code>	Defines the table
<code><tr></code>	Table row
<code><td></code>	Table data cell
<code><th></code>	Table header cell (bold & centered)
<code>border="1"</code>	Adds a visible border around cells

colspan & rowspan

These attributes allow a cell to span across multiple columns or rows:

- **colspan="2"** — makes a cell span 2 columns horizontally
- **rowspan="2"** — makes a cell span 2 rows vertically

Example from today (student grade table):

```
<th colspan="2">PERCENTAGE %</th>
<td rowspan="2">40%</td>
```

Tables Practised Today

Table	Description
Number Grid	3×3 grid of numbers (15, 30, 45, 60, 90...)
Student Grades	NAME, ROLL NUMBER, PERCENTAGE %, GRADE
colspan/rowspan	Merged cells showing shared grade across students
Timetable	Monday/Tuesday schedule with colspan for long classes
Periodic Table	Full periodic table using colspan & rowspan

2. Periodic Table in HTML

The most advanced table created today was the Periodic Table of Elements (table.html). It demonstrated complex use of both colspan and rowspan to replicate the real periodic table layout.

- First row: H and He separated by **colspan="17"**

- Transition metals block used **rowspan="4"** for spacing
- Lanthanide and Actinide series placed in separate rows at the bottom
- Empty spacer cells used with ** ** to maintain layout

3. HTML Images

Images in HTML are embedded using the self-closing **** tag. We explored three methods of referencing images.

Attribute	Purpose	Example
src	Path/URL to the image	"/photo.jpg" or base64 data
alt	Alternative text if image fails	"Car image"
title	Tooltip shown on hover	"car image"
width / height	Set display dimensions	"500px" / "100px"

Image Source Types Practised

1. Base64 Inline Image — Image data is embedded directly in the HTML using a long base64 string. No external file needed, but makes the HTML file large.

```
src="..."
```

2. Relative Path — Image file sits in the same folder. Simple and clean.

```
src="./download.jpg"
```

3. Missing Image with alt text — When the image file doesn't exist, the **alt** attribute shows fallback text to the user.

```
src="./downloads.jpg" alt="Car image"
```

4. Image Formats (from notes.pdf)

The uploaded PDF covered when to use different image formats:

Format	Best Used For
JPEG	Photos with many colors (nature, people, complex scenes)
GIF / PNG	Images with few colors or large flat-color areas (logos, diagrams, illustrations)

Key insight: Photographs of snow, sky, or grass appear to have flat colors but actually consist of many subtly different shades, making JPEG the better choice over GIF/PNG for such images.

5. Table Head & Table Body

HTML provides semantic grouping tags to separate the header row(s) from the data rows inside a table. This improves readability, accessibility, and allows browsers/CSS to treat each section differently.

The Two Tags

Tag	Full Name	Purpose
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<code><thead></code>	Table Head	Wraps the header row(s) — usually contains <code><th></code> cells
<code><tbody></code>	Table Body	Wraps all the data rows — contains <code><td></code> cells
<code><tfoot></code>	Table Foot	Optional — wraps summary/footer rows (e.g. totals)

Syntax Example

```
<table border="1">
  <thead>
    <tr>
      <th>NAME</th>
      <th>ROLL NUMBER</th>
      <th>PERCENTAGE %</th>
      <th>GRADE</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>Usman</td>
      <td>20</td>
      <td>80%</td>
      <td>A</td>
    </tr>
  </tbody>
</table>
```

Why Use *thead* and *tbody*?

- **Semantic clarity:** Makes your HTML easier to read and understand.
- **Accessibility:** Screen readers use these tags to help users navigate table data.
- **CSS styling:** You can target `thead` and `tbody` separately — e.g. give the header a background color.
- **Browser behaviour:** When printing long tables, the browser can repeat the `thead` on every printed page.
- **JavaScript:** Makes it easy to select and manipulate only header or body rows.

Key Takeaways

- ✓ Use `<th>` for header cells — they appear bold and centered automatically.
- ✓ `colspan` merges horizontally; `rowspan` merges vertically.
- ✓ Complex layouts (like the periodic table) are possible with careful use of both.
- ✓ `<thead>` and `<tbody>` add semantic structure — always use them in real projects.
- ✓ The `alt` attribute is important for accessibility when images fail to load.
- ✓ Base64 images are self-contained but increase file size significantly.
- ✓ Choose JPEG for photos, PNG/GIF for flat-color graphics and logos.